

THE FAMILY OF THE SUN

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THE SOLAR SYSTEM CONSISTS OF the Sun, eight recognized planets, over 140 moons, and countless small bodies such as asteroids and comets. Its inner region contains the Sun and the rocky planets—Mercury, Venus, Earth, and Mars. Beyond this lies a ring of asteroids, called the Main Belt; the gas giant planets Jupiter, Saturn, Uranus, and Neptune;

a region called the Kuiper Belt, which contains icy worlds; and finally a vast cloud of comets. In total, the solar system is about 9.3 trillion miles (15 trillion km) across; the planets occupy a zone extending just 3.25 billion miles (6 billion km) from the Sun.

ORBITS IN THE SOLAR SYSTEM

Most orbits of objects in the solar system have the shapes of ellipses (stretched circles). However, for most of the planets, these ellipses are close to being circular. Only Mercury has an orbit that differs very markedly from being circular. All the planets and nearly all asteroids orbit the Sun in the same direction, which is also the direction in which the Sun spins on its own axis. The orbital period (the time it takes to orbit the Sun) increases with distance from the Sun, from 88 Earth days for Mercury to nearly 250 Earth years for Pluto, following a mathematical relationship first discovered by the German astronomer Johannes Kepler in the early 17th century (see p.68). As well as having longer orbits to complete, the planets farther from the Sun move much more slowly.

JUPITER

Orbits the Sun in 11.86 Earth years at an average distance of 483.4 million miles (778.4 million km)

URANUS

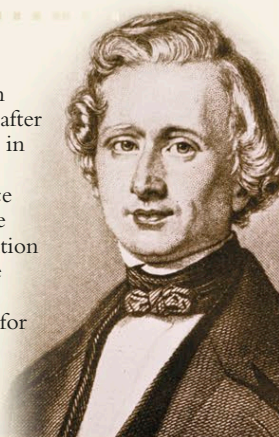
Orbits the Sun in 84.01 Earth years at an average distance of 1.8 billion miles (2.9 billion km)

PLANET ORBITS

All the orbits of the planets, and the Asteroid Belt, lie roughly in a flat plane known as the ecliptic plane. Only Mercury and the dwarf planet Pluto orbit at significant angles to this plane (7.0° and 17.1°, respectively). The planets and their orbits are not shown to scale here.

URBAIN LE VERRIER

Urbain Le Verrier (1811–1877) was a French mathematician and astronomer who, after studying irregularities in the orbit of Uranus, predicted the existence of the planet Neptune and calculated its position in 1846. He asked the German astronomer Johann Galle to look for Neptune, and within an hour, the planet had been found.



THE SUN

The Sun's diameter at the equator is 864,900 miles (1.4 million km), and its equatorial rotation period is about 25 Earth days

EARTH

Orbits the Sun in 365.26 Earth days at an average distance of 92.9 million miles (149.6 million km)

MERCURY

Orbits the Sun in 88 Earth days at an average distance of 36 million miles (57.9 million km)